

Introduction to data analytics: Summary and Highlights

In this lesson, you have learned the following information:

A modern data ecosystem includes a network of interconnected and continually evolving entities that include:

- Data that is available in a host of different formats, structure, and sources.
- Enterprise Data Environment in which raw data is staged so it can be organized, cleaned, and optimized for use by end-users.
- End-users such as business stakeholders, analysts, and programmers who consume data for various purposes.

Emerging technologies such as Cloud Computing, Machine Learning, and Big Data, are continually reshaping the data ecosystem and the possibilities it offers. Data Engineers, Data Analysts, Data Scientists, Business Analysts, and Business Intelligence Analysts, all play a vital role in the ecosystem for deriving insights and business results from data.

Based on the goals and outcomes that need to be achieved, there are four primary types of Data Analysis:

- Descriptive Analytics, that helps decode “What happened.”
- Diagnostic Analytics, that helps us understand “Why it happened.”
- Predictive Analytics, that analyzes historical data and trends to suggest “What will happen next.”
- Prescriptive Analytics, that prescribes “What should be done next.”

The Data Analysis process involves:

- Developing an understanding of the problem and the desired outcome.
- Setting a clear metric for evaluating outcomes.
- Gathering, cleaning, analyzing, and mining data to interpret results.
- Communicating the findings in ways that impact decision-making.

Copy of Data Analytics vs. Data Analysis

The terms **Data Analysis** and **Data Analytics** are often used interchangeably, including in this course.

However it is important to note that there is a subtle difference between the terms and meaning of the words *Analysis* and *Analytics*. In fact some people go far as saying that these terms mean different things and should not be used interchangeably. Yes, there is a technical difference...

The dictionary meanings are:

Analysis - detailed examination of the elements or structure of something

Analytics - the systematic computational analysis of data or statistics

Analysis can be done without numbers or data, such as business analysis psycho analysis, etc. Whereas *Analytics*, even when used without the prefix "Data", almost invariably implies use of data for performing numerical manipulation and inference.

Some experts even say that *Data Analysis* is based on inferences based on historical data whereas *Data Analytics* is for predicting future performance. The design team of this course does not subscribe to this view, and you will see why later in the course as you become familiar with the terms like *predictive analytics*, *prescriptive analytics*, etc.

So in this course we take a more liberal view, and use the terms Data Analysis and Data Analytics to mean the same thing. For example, an earlier video is titled *Defining Data Analysis*, whereas the preceeding video with the viewpoints of several data professionals is titled *What is Data Analytics*. The difference in these titles is not intentional.

Generative AI: An essential Skill for today's Data Analysts

Introduction

As a beginner in data analytics, you're stepping into a field that's rapidly evolving. Generative AI is becoming an essential tool for data analysts, allowing them to create new content and gain deeper insights. Let's explore what generative AI is and how it can enhance your skills.

What is generative AI?

Generative AI refers to a class of artificial intelligence models that create new content such as text, images, music, and more by learning patterns from existing data.

Generative AI can respond naturally to human conversation and serve as a tool for customer service and personalization of customer workflows. For example, you can use AI-powered chatbots, voice bots, and virtual assistants that respond more accurately to customers for first-contact resolution.

How does generative AI work?

Generative AI starts with a prompt that could be in the form of a text, an image, a video, a design, musical notes, or any input that the AI system can process. Various AI algorithms then return new content in response to the prompt. Content can include essays, solutions to problems, or

[realistic fakes](#)

created from pictures or audio of a person.

Early versions of generative AI required submitting data via an API or an otherwise complicated process. Developers had to familiarize themselves with special tools and write applications using languages such as Python.

Now, pioneers in generative AI are developing better user experiences that let you describe a request in plain language. After an initial response, you can also customize the results with feedback about the style, tone, and other elements you want the generated content to reflect.

Key techniques in generative AI:

Generative adversarial networks (GANs): GANs consist of two neural networks: the generator and the discriminator. The generator creates new data, whereas the discriminator evaluates it. Over time, the generator improves to produce realistic data.

Variational autoencoders (VAEs): VAEs encode input data into a compressed format and then decode it back, generating new data points similar to the input data.

Transformers: Used primarily in natural language processing (NLP), transformers generate human-like text by predicting the next word in a sequence. Generative Pre-trained Transformer 3 (GPT-3) is a notable example.

Generative AI models

Generative AI models combine various AI algorithms to represent and process content. For example, to generate text, various NLP techniques transform raw characters (e.g., letters, punctuation, and words) into sentences, parts of speech, entities, and actions, which are represented as vectors using multiple encoding techniques. Similarly, images are transformed into various visual elements, also expressed as vectors. One caution is that these techniques can also encode the biases, racism, deception, and puffery contained in the training data.

Once developers settle on a way to represent the world, they apply a particular neural network to generate new content in response to a query or prompt. Techniques such as GANs and VAEs—neural networks with a decoder and encoder—are suitable for generating realistic human faces, synthetic data for AI training, or even facsimiles of particular humans.

Recent progress in transformers, such as Google's Bidirectional Encoder Representations from Transformers (

[BERT](#)

), OpenAI's

[GPT](#)

, and Google AlphaFold, have also resulted in neural networks that can not only encode language, images, and proteins but also generate new content.

What are the use cases for generative AI?

Generative AI can be applied in various use cases to generate virtually any kind of content. The technology is becoming more accessible to users of all kinds thanks to cutting-edge breakthroughs like GPT that can be tuned for different applications.

Some of the use cases for generative AI include the following:

- Implementing chatbots for customer service and technical support.
- Deploying deepfakes for mimicking people or even specific individuals.
- Improving dubbing for movies and educational content in different languages.
- Writing email responses, dating profiles, resumes, and term papers.
- Creating photorealistic art in a particular style.
- Improving product demonstration videos.
- Suggesting new drug compounds to test.
- [Designing physical products](#)

and buildings.

- Optimizing new chip designs.
- Writing music in a specific style or tone.

What are the benefits of generative AI?

Generative AI can be applied extensively across many areas of the business. It can make it easier to interpret and understand existing content and automatically create new content. Developers are exploring ways that generative AI can improve existing workflows, with an eye to adapting workflows entirely to take advantage of the technology. Some of the potential benefits of implementing generative AI include the following:

- Automating the manual process of writing content.
- Reducing the effort of responding to emails.
- Improving the response to specific technical queries.
- Creating realistic representations of people.
- Summarizing complex information into a coherent narrative.
- Simplifying the process of creating content in a particular style

What are the limitations of generative AI?

Early implementations of generative AI vividly illustrate its many limitations. Some of the challenges generative AI presents result from the specific approaches used to implement particular use cases. For example, a summary of a complex topic is easier to read than an explanation that includes various sources supporting key points. The readability of the summary, however, comes at the expense of a user being able to vet where the information comes from.

Here are some of the limitations to consider when implementing or using a generative AI app:

- It does not always identify the source of content.
- It can be challenging to assess the bias of original sources.
- Realistic-sounding content makes it harder to identify inaccurate information.
- It can be difficult to understand how to tune in to new circumstances.
- Results can gloss over bias, prejudice, and hatred.

What are the concerns surrounding generative AI?

The rise of generative AI is also fueling various concerns. These relate to the quality of results, the potential for misuse and abuse, and the potential to disrupt existing business models. Here are some of the specific types of problematic issues posed by the current state of generative AI:

- It can provide inaccurate and misleading information.
- It is more difficult to trust without knowing the source and provenance of information.
- It can promote new kinds of plagiarism that ignore the rights of content creators and artists of original content.
- It might disrupt existing business models built around search engine optimization and advertising.
- It makes it easier to generate fake news.
- It makes it easier to claim that real photographic evidence of wrongdoing was just an AI-generated fake.
- It could impersonate people for more effective social engineering cyberattacks.
- Given the newness of GenAI tools and their rapid adoption, enterprises should prepare for the inevitable “trough of disillusionment” that’s part and parcel of emerging technology by adopting sound AI engineering practices and making responsible AI a cornerstone of their GenAI efforts, ensuring transparency, ethical considerations, and long-term sustainability in their AI implementations.

What are some examples of generative AI tools?

Generative AI tools exist for various modalities, such as text, imagery, music, code, and voices. Some popular AI content generators to explore include the following:

- Text generation tools include GPT, Jasper, AI-Writer, and Lex.
- Image generation tools include Dall-E 2, Midjourney, and Stable Diffusion.
- Music generation tools include Amper, Dadabots, and MuseNet.
- Code generation tools include codeStarter, Codex, [GitHub Copilot](#), and Tabnine.
- Voice synthesis tools include Descript, Listnr, and Podcast.ai.
- AI chip design [tool](#)

companies include Synopsys, Cadence, Google, and NVIDIA.

Applications of generative AI in data analytics

Generative AI has many applications that can enhance your data analytics work:

Data augmentation: Create synthetic data to augment existing data sets, which is especially useful when data is scarce or imbalanced. This can improve predictive model performance.

Anomaly Detection: Identify anomalies or outliers by understanding the distribution of normal data. This is valuable in fraud detection, network security, and quality control.

Text and image generation: Generate realistic text and images for marketing, content creation, and customer engagement, such as automatic product descriptions and marketing visuals.

Simulation and forecasting: Simulate scenarios and forecast future events by generating potential outcomes from historical data. This is crucial in financial planning, supply chain management, and strategic decision-making.

Conclusion

Generative AI is a transformative technology that can significantly enhance your capabilities as a data analyst. By mastering generative AI techniques, you can unlock new possibilities in data augmentation, anomaly detection, content creation, and forecasting. As you embark on this journey, remember to balance innovation with ethical responsibility, ensuring that AI is used positively.

- **Transformers:** Transformers are primarily used in natural language processing to generate human-like text by predicting the next word in a sequence.
- **Generative Adversarial Networks (GANs):** GANs consist of two neural networks, a generator that creates new data and a discriminator that evaluates it, improving the realism of generated data over time.
- **Generative AI:** Generative AI refers to artificial intelligence models that create new content by learning patterns from existing data.
- **Applications in Data Analytics:** Generative AI can enhance data analytics through data augmentation, anomaly detection, and content generation.
- **Variational Autoencoders (VAEs):** VAEs encode input data into a compressed format and decode it back, generating new data points similar to the input data.

This content was generated by AI, so please check for any mistakes.

Summary and Highlights

In this lesson, you have learned the following information:

The role of a Data Analyst spans across:

- Acquiring data that best serves the use case.
- Preparing and analyzing data to understand what it represents.
- Interpreting and effectively communicating the message to stakeholders who need to act on the findings.
- Ensuring that the process is documented for future reference and repeatability.

In order to play this role successfully, Data Analysts need a mix of technical, functional, and soft skills.

- Technical Skills include varying levels of proficiency in using spreadsheets, statistical tools, visualization tools, programming and querying languages, and the ability to work with different types of data repositories and big data platforms.
- An understanding of Statistics, Analytical techniques, problem-solving, the ability to probe a situation from multiple perspectives, data visualization, and project management skills – all of which come under Functional Skills a Data Analyst needs in order to play an effective role.
- Soft Skills include the ability to work collaboratively, communicate effectively, tell a compelling story with data, and garner support and buy-in from stakeholders. Curiosity to explore different pathways and intuition that helps to give a sense of the future based on past experiences are also essential skills for being a good Data Analyst.

The Data Ecosystem:

Query language: SQL. Accessing and manipulating data in a database.

Program language: Python, R, Java. Developing applications and controlling application behavior.

Shell and Scripting language: ideal for repetitive and time-consuming operational tasks (Unix/Linux Shell, PowerShell)

SQL- structured query language. Accessing and manipulating data. Insert update and delete records, write stored procedures.

Python: high level programming language. Express concepts in fewer lines of code.

Summary and Highlights

In this lesson, you have learned the following information:

A data analyst ecosystem includes the infrastructure, software, tools, frameworks, and processes used to gather, clean, analyze, mine, and visualize data.

Based on how well-defined the structure of the data is, data can be categorized as:

- Structured Data, that is data which is well organized in formats that can be stored in databases.
- Semi-Structured Data, that is data which is partially organized and partially free form.
- Unstructured Data, that is data which can not be organized conventionally into rows and columns.

Data comes in a wide-ranging variety of file formats, such as delimited text files, spreadsheets, XML, PDF, and JSON, each with its own list of benefits and limitations of use.

Data is extracted from multiple data sources, ranging from relational and non-relational databases to APIs, web services, data streams, social platforms, and sensor devices.

Once the data is identified and gathered from different sources, it needs to be staged in a data repository so that it can be prepared for analysis. The type, format, and sources of data influence the type of data repository that can be used.

Data professionals need a host of languages that can help them extract, prepare, and analyze data. These can be classified as:

- Querying languages, such as SQL, used for accessing and manipulating data from databases.
- Programming languages such as Python, R, and Java, for developing applications and controlling application behavior.
- Shell and Scripting languages, such as Unix/Linux Shell, and PowerShell, for automating repetitive operational tasks.

Relational database: SQL. Can use flat files to hold data. Keeps in columns and row formats.

Non-relational database: Data can be stored in free-form fashion or schema-less fashion. Used for processing big data.

Data warehouse goes through extract, transform and load (ETL) process of a large set of data into one data set.

Big Data stores take distributed computational and storage infrastructure to store, scale, and process very large data sets.

Data repositories help to isolate data and make reporting and analytics more efficient and credible while also serving as a data archive.

A relational database (RDBMS) is a collection of data organized into a table structure, where tables can be linked or related based on data common to

each/. Use structured query language or SQL to query data. Are ideal for optimized storage, retrieval and processing of data from large volumes of data. Can retrieve millions of records in seconds.

Well suited for data warehouses where can be optimized for online analytical processing (OLAP) or IoT solutions providing the speed and ability to collect and process data from edge devices.

Limitations: Does not work well with semi-structured and unstructured data. Migration is possible only when source and destination tables have identical schemas and data types. Entering a value greater than the defined length of a data field results in loss of data.

Predominantly used for structured data.

NoSQL (not only SQL/non-SQL) is a non-relational database design that provides flexible schemas for storage and retrieval of data. Do not use traditional row/column/table database. Don't use query language to probe.

Four types:

Key-value store: data in a key-value database is stored as a collection of key-value pairs. A key represents an attribute of the data and is an identifier. User session data or user preferences are represented in it.

Not good if want to query data on specific data value, need relationships between data values, or need multiple unique keys

Document based: store each record and associated data within a single document. Enable flexible indexing, powerful ad hoc queries, and analytics over collection of documents. Preferred for eCommerce platforms, medical records storage, CRM platforms, and analytics platforms.

Not good if want to run complex search queries or perform multi-operation transactions

Column based: stored in cells grouped as columns of data instead of rows, logical grouping of columns is called a family. All cells corresponding to a column are saved as a continuous disk entry, making access and search easier and faster. Great for systems that require heavy write requests, storing time-series data, weather data, and IoT data.

Not good if need to run complex queries or change querying patterns frequently.

Graph based: store data with graphical model to represent and store data. Excellent with working with data with lot of interconnections. Social networks, fraud management, etc.

Not good if want to process high volumes of transactions

NoSQL advantages: handle large volumes of structured, semi-structured and unstructured data. Run as distributed system scaled across multiple data centers. An efficient and cost-effective scale-out architecture that provides additional capacity and performance with the addition of new nodes. Simple design, better control over availability, and improved scalability that makes it agile, flexible, and support quick iterations.

RDBMS schemas rigidly define how all data are inserted into databases and must be composed of types whereas NoSQL can be schema-agnostic, allowing unstructured and semi-structured data to be stored and manipulated. RDBMS maintain high-end, commercial relational database management systems that can be expensive but NoSQL are designed for low-cost commodity hardware. RDBMS support ACID-compliance, which ensures reliability of transactions and crash recovery but most NoSQL databases are not ACID compliant. RDBMS are more mature and well-documented whereas NoSQL are relatively newer technology.

Data warehouse serves as “single source of truth”, multipurpose enabler for operations and analytics. A data mart is a small subsection of the data warehouse that provides analytical capabilities for a restricted area of the data warehouse. It offers isolated security and isolated performance. The most important role is business specific reporting and analytics.

Data lake is a storage repository able to store data in unprocessed form, structured or unstructured or semi-structured. Pool of raw data where each data element is given a unique identifier and is tagged with metadata for further use. Retains all source data without exclusions. Selected and organized based on use case needed for. Sometimes used as staging area for data warehouse. Most important role is predictive and advanced analytics.

Extract, transform, and load. Is automated process where gather raw data, extract info needed, clean standardize and transform data into usable format, and load data into data repository. Extraction can be through batch processing (large chunks of data), stream processing (data pulled in real-time from source, transformed in transit, and loaded into data repository). Transformation standardizes data, removes duplicates filtering out unneeded data. Loading is the transportation of processed data into data repository. Can be initial (population all of data), incremental (applying ongoing updates and modifications as needed), or full refresh (enables refreshing of data). Load verification includes checks for missing data, etc.

A data pipeline encompasses entire journey of moving data from one system to another including ETL, can be used for both batch and streaming data, supports long-running batch queries and smaller interactive queries. Typically

loads data into a data lake or a variety of target destinations including others applications and visualization tools.

Foundations of big data:

V's of big data:

Velocity: speed at which data accumulates

Volume: scale of data or increase in amount of data stored

Variety: diversity of data

Veracity: quality and origin of data

Value: ability and need to turn data into value (medical or social benefits, etc.)

Big data processing tools:

Apache Hadoop (distributed file system): collection of tools that provides distributed storage and processing of data. Has nodes (single computer) and multiple nodes form clusters. Stores data with no format requirements. Can do structured and non-structured. Hadoop distributed file system (HDFS), is a storage system for big data that runs on multiple commodity hardware connected through a network. Partitioning through nodes and can scale to hundreds of nodes in a single cluster.

Apache Hive: a data warehouse for data query and analysis built on top of Hadoop. Reads, writes, and manages large data set files stored in either HDFS or other data storage systems. HIVE is not good for something needing fast response time, high percentage of write operations. Better suited for data warehousing tasks such as ETL, reporting and data analysis. Includes tools with easy access to SQL

Spark: a distributed analytics framework for complex data analytics in real time. Data processing engine designed to extract and process large volumes of data: interactive analytics, streams processing, machine learning, data integration, and ETL. Takes advantage of in memory processing with significant increases speed of computations, provides interfaces for major programming languages, can run using standalone clustering tech, can run on top of other infrastructure such as Hadoop, can access data in large variety of data sources, including HDFS and hive, processes streaming data fast and performs analytics in real time.

Summary and Highlights

In this lesson, you have learned the following information:

A Data Repository is a general term that refers to data that has been collected, organized, and isolated so that it can be used for reporting, analytics, and also for archival purposes.

The different types of Data Repositories include:

- Databases, which can be relational or non-relational, each following a set of organizational principles, the types of data they can store, and the tools that can be used to query, organize, and retrieve data.
- Data Warehouses, that consolidate incoming data into one comprehensive storehouse.
- Data Marts, that are essentially sub-sections of a data warehouse, built to isolate data for a particular business function or use case.
- Data Lakes, that serve as storage repositories for large amounts of structured, semi-structured, and unstructured data in their native format.
- Big Data Stores, that provide distributed computational and storage infrastructure to store, scale, and process very large data sets.

ETL, or Extract Transform and Load, Process is an automated process that converts raw data into analysis-ready data by:

- Extracting data from source locations.
- Transforming raw data by cleaning, enriching, standardizing, and validating it.
- Loading the processed data into a destination system or data repository.

Data Pipeline, sometimes used interchangeably with ETL, encompasses the entire journey of moving data from the source to a destination data lake or application, using the ETL process.

Big Data refers to the vast amounts of data that is being produced each moment of every day, by people, tools, and machines. The sheer velocity, volume, and variety of data challenge the tools and systems used for conventional data. These challenges led to the emergence of processing tools and platforms designed specifically for Big Data, such as Apache Hadoop, Apache Hive, and Apache Spark.

Module 3

Gathering and Wrangling data

Step 1: Determine information you want to collect: specific information you need and possible sources for this data

Step 2: define a plan for collecting data: establish timeframe for collecting data, how much data is sufficient for analysis?, Define dependencies, risks, and mitigation plan

Step 3: Determine data collection methods: depending on sources of data, type of data, timeframe over which you need the data, and the volume of data

Quality, security, and privacy are key considerations depending on data and how you gather data.

Data needs to be free of errors, accurate, complete, relevant and accessible

Data governance: look out for security, regulation, and compliances. Procedures related to usability, integrity and availability of data.

Data privacy: Needs to include issues such as confidentiality, compliance to mandated regulations, and laws.

Data sources

-Can be internal or external to the organization.

Primary data refers to information obtained directly from the source: organization's CRM, HR, or work groups

Secondary data refers to information retrieved from existing sources: external databases, research articles, etc.

Third party data refers to data purchased from aggregators that collect data from various sources and gather it solely for purposes of selling data

Databases can be a source of primary, secondary and third party data. Web is a source of publicly available data that is available to companies and individuals.

Sensor data produced by wearable devices or smart anythings.

Data exchange is a source of third-party data the involves the voluntary sharing of data providers and data consumers.

Surveys gather information through questionnaires distributed to select groups.

Census data is popularly used for population dynamics.

Interviews are source for gathering qualitative data such as participant's opinions.

Observation studies include monitoring participants in a specific environment or while performing a task.

How to gather and import data:

SQL or structured query language is a querying language used for extracting information from relational databases:

Offers commands to specify: what to retrieve from database, table from which it needs to be extracted, grouping records with matching values, dictating sequence in which query results are displayed, limited number of results that can be returned by the query.

Some non-relational interfaces come with their own querying tools such as CQL for Cassandra and graph QL for Neo4J.

Application programming interfaces (APIs) popularly used for extracting data from variety of dataset sources. Are invoked from applications that require data and access an endpoint containing the data. Endpoints can include databases, web services, and data marketplaces. Also used for data validation.

Web scraping is used for downloading specific data from web pages based on defined parameters. Extract text, contact info, images, videos, etc. Rss feeds are used for capturing updated data.

Data streams are a popular source for aggregating constant streams of data flowing from sources.

Data exchange platforms allow the exchange of data between data providers and data consumers. Have well defined exchange standards and protocols.

Provide data licensing workflows and de-identification and protection of personal information legal frameworks (AWS exchange, snowflake, etc.)

Numerous other sources can be tapped into specific data needs.

Data identified and gathered needs to be imported into data repository.

Specific data repositories are optimized for certain types of data:

Structured data; relational databases store structured data (SQL and noSQL)

semi-structured data can be stored in NoSQL clusters (XML and JSON)

Unstructured data (NoSQL and data lakes)

In this lesson, you have learned:

- The process of identifying data begins by determining the information that needs to be collected, which in turn is determined by the goal you seek to achieve.
- Having identified the data, your next step is to identify the sources from which you will extract the required data and define a plan for data collection. Decisions regarding the timeframe over which you need your data set, and how much data would suffice for arriving at a credible analysis also weigh in at this stage.
- Data Sources can be internal or external to the organization, and they can be primary, secondary, or third-party, depending on whether you are obtaining the data directly from the original source, retrieving it from externally available data sources, or purchasing it from data aggregators.
- Some of the data sources from which you could be gathering data include databases, the web, social media, interactive platforms, sensor devices, data exchanges, surveys and observation studies.
- Data that has been identified and gathered from the various data sources is combined using a variety of tools and methods to provide a single interface using which data can be queried and manipulated.
- The data you identify, the source of that data, and the practices you employ for gathering the data have implications for quality, security, and privacy, which need to be considered at this stage.

What is Data wrangling:

An iterative process that involves data exploration, transformation, validation, and making it available for a credible and meaningful analysis

Data wrangling process:

4 steps:

Discovery, transformation, validation, and publishing

Discovery is about understanding data for your use case, how can best clean structure and organize data.

Transformation is bulk of wrangling. Is tasks to approach data such as structuring data, normalizing and denormalizing data, cleaning data, and enriching data.

Structuring data: need to change form or schema of data. Joins and unions are most common.

Joins combine columns.

Unions combine rows.

Can also include normalization and denormalization.

Clean unused data, reducing redundance and inconsistency.

Denormalization data includes combining data into a single table for faster access to data.

Cleaning data- fix irregularities in data to produce more accurate data, missing data, incomplete data, etc.

Enriching data: adding data points that make your data more meaningful

Validation: check the quality of data after structuring, normalizing, denormalizing, cleaning , and enriching of data.

Validation is repetitive steps to verify consistency, quality, and security of data.

Publishing: delivering the output of the wrangled data for downstream project needs.

Critical for documentation. All phases of data wrangling are iterative in nature. Need to document all considerations and actions.

A. Tools for data wrangling

A. Excel power query/spreadsheets

B. OpenRefine

C. Google Dataprep

D. Watson

E. Trifacta

F. Python

G. R

B. Spreadsheets

1. Open refine is an open-source tool that can import and export data in wide variety of formats.
2. Can learn data, transform from one data format to another and extend data.

Google data prep is an intelligent data service that can visually explore, clean and prepare both structured and unstructured data

Watson studio refinery allows you to discover, cleanse, and transform data with built-in operations. Detects data types and classifications automatically

Trifacta wrangler is an interactive cloud-based service for cleaning and transforming data.

Python has a huge library and set of packages that offer powerful data manipulation capabilities.

Numpy is the most basic package that python offers. Provides support for late multi-dimensional arrays and matrices.

Pandas is designed for fast and easy data analysis operations. Allows complex operations such as merging, joining, and transforming huge datasets.

R also offers a series of libraries and packages that are explicitly created for wrangling data.

Dplyr: a powerful library for data wrangling with precise and straightforward syntax, table, and JSON wrangler.

Decision on best tool for needs depends on data and other information.

Data cleaning:

Forms a very significant part of transformation part of data wrangling work blow (inspection, cleaning, and verification)

Inspection includes detecting issues and errors.

Data profiling allows you to understand source data structure, content, and inter-relationships. Visualizing data can spot outliers.

Cleaning of data technique will depend on data and use cases.

- Missing values can cause unexpected or biased results: can source the data, remove data with missing info, or impute.

- Duplicate data need to be removed

- Data type conversion

- Standardizing data

- Syntax errors such as white spaces, extra spaces, typos, and formats that need to be fixed

- Outliers: depending on use case may need to remove if doesn't serve use case

Verification includes inspecting results to establish effectiveness and accuracy achieved as a result of data.

DOCUMENT changes and reasons for undertaking changes and quality of currently stored data.

Summary and Highlights

In this lesson, you have learned the following information:

Once the data you identified is gathered and imported, your next step is to make it analysis-ready. This is where the process of Data Wrangling, or Data Munging, comes in. Data Wrangling is an iterative process that involves data exploration, transformation, and validation.

Transformation of raw data includes the tasks you undertake to:

- Structurally manipulate and combine the data using Joins and Unions.

- Normalize data, that is, clean the database of unused and redundant data.

- Denormalize data, that is, combine data from multiple tables into a single table so that it can be queried faster.

Clean data, which involves profiling data to uncover quality issues, visualizing data to spot outliers, and fixing issues such as missing values, duplicate data, irrelevant data, inconsistent formats, syntax errors, and outliers.

Enrich data, which involves considering additional data points that could add value to the existing data set and lead to a more meaningful analysis.

A variety of software and tools are available for the Data Wrangling process. Some of the popularly used ones include Excel Power Query, Spreadsheets, OpenRefine, Google DataPrep, Watson Studio Refinery, Trifacta Wrangler, Python, and R, each with their own set of characteristics, strengths, limitations, and applications.

Mining & Visualizing data and communication results

I. Statistical analysis

- A. Application of statistical methods to a sample of data to develop and understanding of what that data represents
 - 1. Sample- a representative selection drawings from a total population
 - 2. Population- a discrete group of people of things that can be identified by at least one attribute
- B. Helps ensure data is interpreted correctly and apparent relationships are meaningful
- C. Types
 - 1. Descriptive statistics- summarizing info about the sample
 - a) Enables to present data in meaningful way, simpler interpretation of data, don't need to draw conclusions about population data was drawn from
 - b) Includes central tendency, dispersion, and skewness
 - (1) Central tendency
 - (a) Locate center of data sample (mean, median, mod)
 - (2) Dispersion
 - (a) Measure of variability in dataset (variance, SD, and range)
 - (i) Variance is how far away data points fall from center
 - (ii) Lower variability is more consistent, higher variability means dissimilar values with likelihood of extreme values
 - (iii) Standard deviation tells you how tightly data is clustered around the mean
 - (iv) Range is distance between smallest and largest values
 - (b) Skewness is the measure of whether the distribution of values is symmetrical around a central value or skewed left or right

- (i) Can affect which types of analyses are valid to perform
- 2. Inferential statistics- make inferences or generalizations about the broader population
 - a) Takes data from sample to make inferences about larger population from which the sample was drawn.
 - b) Helps generalize
 - c) Common methods:
 - (1) Hypothesis testing- can tell you whether efficacy of vaccine observed in control group is likely to occur in population as well
 - (2) Confidence intervals incorporate the uncertainty and sample error to create a range of values the actual population value is likely to fall within
 - (3) Regression analysis incorporates hypothesis tests that help determine whether the relationships observed in the sample data actually exist in the population rather than just the sample
- D. Packages
 - 1. SAS, SPSS, and statsoft
- E. Statistics combined with data mining, together help in better decision making. Provide measures and methodologies necessary for data mining and identify patterns that help identify differences between random noise and significant findings.

What is Data mining?

- I. Process of extracting knowledge from data
- II. Use of pattern recognition technologies, statistical analysis, and mathematical techniques
- III. Identify correlations and understand patterns and trends
- IV. Patterns and trends
 - A. Pattern recognition is the discovery of regularities or commonalities in data
 - B. Trend is general tendency of a set of data to change over time
- V. Applications
 - A. Profiling customer behaviors, needs, and campaigns for commercials
 - B. Prioritize treatment
 - C. Performance data of students
 - D. Deploy police where crime is higher
- VI. Data mining techniques
 - A. Classification
 - 1. Classifies attributes into target categories
 - B. Cluster
 - 1. Grouping data into clusters so can be treated as groups
 - C. Anomaly or outlier detection
 - 1. Find patterns in data that are not normal or unexpected

- D. Association rule mining
 - 1. Establish relationship between two data events
 - E. Sequential patterns
 - 1. Trace series of events that take place in a sequence
 - F. Affinity grouping
 - 1. Discovering co-occurrence in relationships
 - G. Decision trees
 - 1. Build classification models in form of a tree structure with multiple branches, each representing a probable occurrence
 - H. Regression
 - 1. Identify nature of relationship between two variables, which could be causal or correlational
- VII. Data mining helps separate noise from real occurrences.

Tools for data mining

- I. Spreadsheets are commonly used to perform basic data mining tasks
 - A. Create pivot tables
 - B. Draw comparisons from different sets of data
 - C. Add-ins
 - 1. Data mining client, exlminer, knowledgeminer
 - a) Allows you to perform mining tasks such as classification, regression, association rules, clustering, and model building
 - 2. Text analysis, text mining, and google analytics available in google sheets
- II. R-language
 - A. One of the most widely used languages for performing statistical modeling and computations by statisticians and data miners
 - B. R libraries can perform data mining operations such as:
 - 1. Regression, classification, data clustering, text mining, etc.
 - C. TM is a framework for text mining application within R
 - D. twitteR is a framework for mining tweets
- III. Python
 - A. Pandas is an open-source module for working with data structures and analysis
 - 1. Upload data in any format and can manipulate data
 - 2. Perform basic numerical computations such as mean, median, mode, and range
 - 3. Calculate stats and answer questions regarding correlation between data and distribution of data
 - 4. Explore data visually and quantitatively
 - B. Numpy
 - 1. A tool for mathematical computing and data preparation in python
 - 2. Offers built-in functions and capabilities for data mining
 - C. Jupyter
 - 1. Tool of choice with python to do stat analysis and data mining in python
- IV. Statistical package for social sciences (SPSS)

- A. Used for advanced analytics, text analytics, trend analysis, validation of assumptions, and translation of business problems into data science solutions
 - B. Is closed source
 - C. License for use
 - D. Has easy to use interface with minimal coding
 - E. Efficient data management tools
 - F. Popular for indepth analysis tools
- V. IBM watson studio
 - A. Included in IBM cloud Pak for Data, leveraging open and closed source tools
 - 1. Available through web browser, cloud, private cloud, and desktop app
 - 2. Allows collaboration of projects
 - 3. Includes SPSS modeler flows for predictive models for data
- VI. SAS enterprise miner
 - A. Comprehensive, graphical workbench for data mining
 - B. Provides capabilities for data exploration (relationships between data)
 - C. Manage information from various sources, mine, and transform data, and analyze statistics
 - D. Graphical user interface for non-technical users.
 - E. You can
 - 1. Identify patterns in data using range of available modeling techniques
 - 2. Explore relationships and anomalies in data
 - 3. Analyze big data
 - 4. Validate reliability of findings from data analysis process
 - 5. Can handle large amounts of data
- VII. Considerations for what to use
 - A. Data size and structures supported by tool
 - B. Key features
 - C. Data visualization capabilities
 - D. Infrastructure needs
 - E. Ease of use
 - F. Learnability

Summary and Highlights

In this lesson, you have learned the following information:

Statistics is a branch of mathematics dealing with the collection, analysis, interpretation, and presentation of numerical or quantitative data.

Statistical Analysis involves the use of statistical methods in order to develop an understanding of what the data represents.

Statistical Analysis can be:

- Descriptive; that which provides a summary of what the data represents.
Common measures include Central Tendency, Dispersion, and Skewness.
- Inferential; that which involves making inferences, or generalizations, about data.
Common measures include Hypothesis Testing, Confidence Intervals, and Regression Analysis.

Data Mining, simply put, is the process of extracting knowledge from data. It involves the use of pattern recognition technologies, statistical analysis, and mathematical techniques, in order to identify correlations, patterns, variations, and trends in data.

There are several techniques that can help mine data, such as, classifying attributes of data, clustering data into groups, establishing relationships between events, variables, and input and output.

A variety of software and tools are available for analyzing and mining data. Some of the popularly used ones include Spreadsheets, R-Language, Python, IBM SPSS Statistics, IBM Watson Studio, and SAS, each with their own set of characteristics, strengths, limitations, and applications.

Overview of communicating and sharing data analysis findings

- I. Success of communication depends on how well others can understand and trust your insights to take further action
 - A. Who is my audience?
 - B. What is important to them?
 - C. What will help them trust me?

Introduction to data visualization

- I. Data visualization is the discipline of communicating information through the use of visual elements such as graphs, charts, and maps. Its goal is to make information easy to comprehend, interpret and retain
- II. Have to choose visualization that effectively delivers your findings to your audience

- A. What is relationship I am trying to establish?
- B. Do I want to compare multiple values?
- C. Do I want my audience to see correlation between two variables?
- D. Do I want to detect anomalies?
- E. What is the question I'm trying to answer?
 - 1. Need to know this for audience and every dataset and information you visualize
 - 2. What should be key takeaways, what does audience need to know, what are questions they may have?

III. Common graph types

- A. Bar charts are great for comparing related data sets or parts of a whole
- B. Column charts compare values side-by-side, you can use them to show change over time
- C. Pie charts show the breakdown of an entity into its sub-parts and the proportion of the sub-parts in relation to one another. Each portion of pie represents a static value or category
- D. Line charts display trends. Great for showing how a data value is changing in relation to a continuous variable

IV. Dashboards

- A. Organize and display reports and visualizations coming from multiple data sources into a single graphical interface
 - 1. Can present operational and analytical data
- B. Birds eye view of complete picture while allowing to drill down to next level of info for each parameter
 - 1. Easy to comprehend
 - 2. Collaboration easy
 - 3. Allow to generate reports on the go

Data visualization software and tools

I. Most common

- A. Spreadsheets
 - 1. Most commonly used for graphical representations of data sets
 - 2. Provides several chart types including Gantt charts, waterfall charts, and combo charts
 - 3. Provides recommendations on visual representation
 - 4. Can add chart title, change colors, and add labels
 - 5.
- B. Jupyter notebook and python libraries
 - 1. Open source web application that provides a way to explore data and create visualizations
 - 2. Python provides host of libraries that are used for data visualization
 - a) Matplotlib
 - (1) Widely used python data vis library
 - (2) Provides different kinds of 2D and 3D plots and flexibility to create plots in different ways

- (3) Creates high-quality interactive graphs and tools with a few lines of code

- (4) Has a large community support and cross-platform

- b) Bokeh

- (1) Provides interactive charts and plots

- (2) Delivers high-performance interactivity over large or streaming datasets

- (3) Flexibility for applying interaction, layouts, and different styling options to visualization

- (4) Transform visualizations written in other Python libraries, such as matplotlib, seaborn, and ggplot

- c) Dash

- (1) A python framework for creating interactive web-based visualizations

- (2) Build highly interactive web applications using python code

- (3) Does not require knowledge of HTML and javascript

- C. R-studio and R-shiny

- 1. Basic visualizations such as histograms, bar charts, line charts, box plots, and scatter plots

- 2. Advanced visualizations such as heat maps, mosaic maps, 3D graphs and correlograms

- 3. Shiny

- a) R package that helps build interactive web apps that can be hosted as standalone apps on a webpage

- (1) Can also build dashboards using Shiny

- (2) Ease of working with it is what popularized it

- D. IBM cognos analytics

- 1. End to end analytics solution

- a) Importing custom visualizations

- b) Forecasting feature that provides time-series data modeling and forecasts

- c) Recommendation for visualization based on your data

- d) Conditional formatting which allow you to see distributing of data and highlight exceptional data points

- e) Known for superior visualization and overlaying using geospatial capability

- E. Tableau

- 1. Software company that produces interactive data visualization products

- a) Create interactive graphs and charts in form of dashboards and worksheets, with drag and drop gestures

- b) Publish results in form of stories

- c) Import R and python scripts

- d) Compatible with

- (1) Excel

- (2) Text
- (3) Relational databases
- (4) Cloud databases

F. Microsoft Power BI

1. Cloud-based business analytics service from Microsoft that enables you to create reports and dashboards
 - a) Powerful and flexible tool known for speed and efficiency
 - b) Drag and drop interface
 - c) Compatible with multiple sources, including Excel, SQL Server and cloud-based data repositories
 - d) Provides ability to collaborate and share dashboards and reports securely
 - e) "Pins" in dashboard

II. Considerations

- A. Ease of use
- B. Purpose of visualization

Visualization tools relied on most by experts

- 1) IBM Cognos
- 2) Looker and Tableau
- 3) Tableau and Power BI
- 4) Microsoft Excel

Summary and Highlights

In this lesson, you have learned the following information:

Data has value through the stories that it tells. In order to communicate your findings impactfully, you need to:

- Ensure that your audience is able to trust you, understand you, and relate to your findings and insights.
- Establish the credibility of your findings.
- Present the data within a structured narrative.
- Support your communication with strong visualizations so that the message is clear and concise, and drives your audience to take action.

Data visualization is the discipline of communicating information through the use of visual elements such as graphs, charts, and maps. The goal of visualizing data is to make information easy to comprehend, interpret, and retain.

For data visualization to be of value, you need to:

- Think about the key takeaway for your audience.
- Anticipate their information needs and questions, and then plan the visualization that delivers your message clearly and impactfully.

There are several types of graphs and charts available for you to be able to plot any kind of data, such as bar charts, column charts, pie charts, and line charts.

You can also use data visualization to build dashboards. Dashboards organize and display reports and visualizations coming from multiple data sources into a single graphical interface. They are easy to comprehend and allow you to generate reports on the go.

When deciding which tools to use for data visualization, you need to consider the ease-of-use and purpose of the visualization. Some of the popularly used tools include Spreadsheets, Jupyter Notebook, Python libraries, R-Studio and R-Shiny, IBM Cognos Analytics, Tableau, and Power BI.

Career opportunities in Data analysis:

12.3% growth of compound annual growth rate.

Data analyst specialist roles and domain specialist roles.

Data analyst specialists go from associate data analyst to principal analyst. Advancing technical, statistical, and analytical skills from foundational to expert level. Tools and platforms, data analysis process, and use cases. Communication skills, stakeholder management skills, project management skills.

Lead or principal analysts may be responsible for establishing processes in your team, making recommendations for software, etc.

Domain specialist roles are seen as an authority in their domain such as HR, healthcare, sales, finance, social media, or digital marketing. Not as skilled.

Analytics enabled job roles: project managers, marketing managers, HR managers. Analytics skills lead to efficiency.

Other data professions: data engineers and data scientists. Business analytics and business intelligence analytics are also open.

What do employers look for in a data analyst?

Honesty, communication, fluency with numbers, AB tests understanding, complex analysis, strong SQL skill, growth mindset, programming skills including python, R, & SQL, detail-oriented, over-achievers, think outside the box, trouble-shooting, problem solving, ability to know data, know what data will solve problems at hand, dynamic and adaptable, pick up technical skills quickly.

The many paths to data analysis

- I. Academic degree
 - A. Online training programs
- II. Mid-career transition
 - A. Research knowledge and skills you need, available job opportunities, and available growth
- III. Explore domain specialist/functional analyst
 - A. Begin with advantage of industry experience and skill yourself with other areas such as statistics and programming

- IV. FOR YOUR CV: Problem-solving, project management, communication and storytelling are important, how you have previously used these skills in other positions?

Career options for data professionals:

- I. Data analyst
- II. Data scientist
- III. Statistician
- IV. Data engineer
- V. BI analyst
- VI. Machine learning or AI engineer
- VII. Business and inform company strategy
- VIII. Manage other data analysts
- IX. Bookkeeper
- X. Accountant
- XI. CPA
- XII. Stockbroker
- XIII. Financial analyst
- XIV. Real estate broker

Build a professional portfolio, use data sets available online, look for opportunities within your existing job

AI for data analytics

- I. Generative IA is a category of AI that focuses on creating new, synthetic data.
- II. Applications
 - A. Testing and development

- B. Enhancing data visualizations
 - C. Filling in missing data points
- III. Creates synthetic data sets and augments existing ones
 - A. Fills in missing data
- IV.
- V. Transform data representation
 - A. Can convert text into images or vice versa
 - B. Automates and enhances data cleaning normalization
 - C. Transform process
 - D. Streamline raw data to processed data
- VI. Querying databases
 - A. Formulates complex queries
 - B. Optimizes database interactions
 - C. Adapts to evolving data structures
 - D. Empowers Q&A models
- VII. Interactive visualizations and storytelling
 - A. Makes data visualizations informative and visually striking
 - B. Makes complex information accessible and engaging
 - 1. Tableau AI, AI assistant of IBM cognos, google's looker AI
 - C. Simplifies and accelerates dashboard creation
 - D. Enhances storytelling in data analytics
- VIII. Future of data analytics
 - A. Automating exploratory data analysis
 - B. Creating captivating data-driven narratives
- IX. Learned:
 - A. Generative AI is a catalyst for innovation in data analytics
 - B. Shaping a future where data is analyzed and created
 - C. Reshapes how you interact and derive meaning from data

Summary and Highlights

In this lesson, you have learned the following information:

Data Analyst roles are sought after in every industry, be it Banking and Finance, Insurance, Healthcare, Retail, or Information Technology.

Currently, the demand for skilled data analysts far outweighs the supply, which means companies are willing to pay a premium to hire skilled data analysts.

Data Analyst job roles can be broadly classified as follows:

- Data Analyst Specialist roles - On this path, you start as a Junior Data Analyst and move up to the level of a Principal Analyst by continually advancing your technical, statistical, and analytical skills from a foundational level to an expert level.
- Domain Specialist roles - These roles are for you if you have acquired specialization in a specific domain and want to work your way up to be seen as an authority in your domain.
- Analytics-enabled job roles - These roles include jobs where having analytic skills can up-level your performance and differentiate you from your peers.
- Other Data Professions - There are several other roles in a modern data ecosystem, such as Data Engineer, Big Data Engineer, Data Scientist, Business Analyst, or Business Intelligence Analyst. If you upskill yourself based on the required skills, you can transition into these roles.

There are several paths you can consider in order to gain entry into the Data Analyst field. These include:

- An academic degree in Data Analytics or disciplines such as Statistics and Computer Science.

- Online multi-course specializations offered by learning platforms such as Coursera, edX, and Udacity.
- Mid-career transition into Data Analysis by upskilling yourself. If you have a technical background, for example, you can focus on developing the technical skills specific to Data Analysis. If you do not have a technical background, you can plan to skill your self in some basic technologies and then work your way up from an entry-level position.

Using Data Analysis for Detecting Credit Card Fraud

Companies today are employing analytical techniques for the early detection of credit card frauds, a key factor in mitigating fraud damage. The most common type of credit card fraud does not involve the physical stealing of the card, but that of credit card credentials, which are then used for online purchases.

Imagine that you have been hired as a Data Analyst to work in the Credit Card Division of a bank. And your first assignment is to join your team in using data analysis for the early detection and mitigation of credit card fraud.

In order to prescribe a way forward, that is, suggest what should be done in order for fraud to get detected early on, you need to understand what a fraudulent transaction looks like. And for that you need to start by looking at historical data.

Here is a sample data set that captures the credit card transaction details for a few users.

IP Address	User ID	Account Number	Age	Shipping Address	Transaction Date	Transaction Time	Transaction Value	Product Category	Units Purchased
3.56.123.0	johnp	25671147	32	1542, Orchid Lane, WA 98706, US	15-5-20	15:00:05	\$121.58	Clothing	1
3.56.123.0	johnp	25671147	32	1542, Orchid Lane, WA 98706, US	10-6-20	10:23:10	\$79.23	Electronics	2
3.56.123.0	johnp	25671147	32	1542, Orchid Lane, WA 98706, US	1-6-20	07:12:45		Home Décor	1
1.186.52.7	johnp	25671147	32	In-store	3-6-20	01:11:10	\$2,009.99	Electronics	10
	johnp	25671147	32	In-store	2020-06-03	01:15:12	\$4,131.00	Electronics	15
1.186.52.7	johnp	25671147	32	P.O. Box 1049	03-06-2020	01:22:24	\$3,010.50	Tools	20
1.58.167.2	davidg	51422789	47	90 Robinson Blvd, Alberta, 97602, Canada	15 May 2020	17:02:08	\$234.20	Furniture	1
1.58.167.2	davidg	51422789	47	90 Robinson Blvd, Alberta, 97602, Canada	18 May 2020	19:12:45	\$141.00	Kithcen Supplies	3
	davidg	51422789	47	90 Robinson Blvd, Alberta, 97602, Canada	01 June 2020	17:34:15	\$157.25	Car Spares	2
1.58.167.2	davidg	51422789	47	90 Robinson Blvd, Alberta, 97602, Canada	13 June 2020	18:02:10	\$59.99	Kithcen Supplies	1
172.165.10.1	ellend	11568528		P.O. Box 1322	07 June 2020	15:53:12	\$99.99	Clothing	1
172.165.10.1	ellend	11568528		P.O. Box 1322	08 June 2020	17:15:30	\$53.15	Beauty	1
1.167.255.10	ellend	11568528		P.O. Box 5401	02 July 2020	00:05:10	\$4,895.00	Laptop	1

Descriptive techniques of analysis, that is, techniques that help you gain an understanding of what happened, include the identification of patterns and anomalies in data. Anomalies signify a variation in a pattern that seems uncharacteristic, or, out of the ordinary. Anomalies may occur for perfectly valid and genuine reasons, but they do warrant an evaluation because they can be a sign of fraudulent activity.

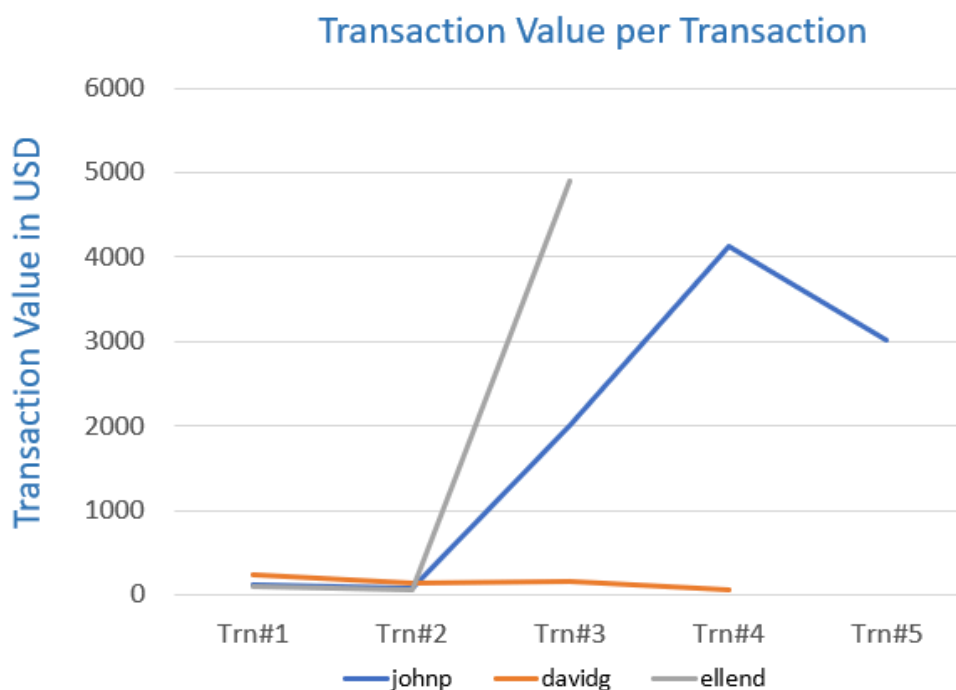
Past studies have suggested that some of the common events that you may need to watch out for include:

- A change in frequency of orders placed, for example, a customer who typically places a couple of orders a month, suddenly makes numerous transactions within a short span of time, sometimes within minutes of the previous order.
- Orders that are significantly higher than a user's average transaction.
- Bulk orders of the same item with slight variations such as color or size—especially if this is atypical of the user's transaction history.
- A sudden change in delivery preference, for example, a change from home or office delivery address to in-store, warehouse, or PO Box delivery.
- A mismatched IP Address, or an IP Address that is not from the general location or area of the billing address.

Before you can analyze the data for patterns and anomalies, you need to:

- Identify and gather all data points that can be of relevance to your use case. For example, the card holder's details, transaction details, delivery details, location, and network are some of the data points that could be explored.
- Clean the data. You need to identify and fix issues in the data that can lead to false or incomplete findings, such as missing data values and incorrect data. You may also need to standardize data formats in some cases, for example, the date fields.

Finally, when you arrive at the findings, you will create appropriate visualizations that communicate your findings to your audience. The graph below samples one such visualization that you would use to capture a trend hidden in the sample data set shared earlier on in the case study.



In the next section you will be asked to answer the following 5 (five) questions based on this case study:

1. List at least 5 (five) data points that are required for the analysis and detection of a credit card fraud. (3 marks)
2. Identify 3 (three) errors/issues that could impact the accuracy of your findings, based on a data table provided. (3 marks)
3. Identify 2 (two) anomalies, or unexpected behaviors, that would lead you to believe the transaction may be suspect, based on a data table provided. (2 marks)
4. Briefly explain your key take-away from the provided data visualization chart. (1 mark)
5. Identify the type of analysis that you are performing when you are analyzing historical credit card data to understand what a fraudulent transaction looks like. [Hint: The four types of Analytics include: Descriptive, Diagnostic, Predictive, Prescriptive] (1 mark)

1.